THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 5: **Life Processes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The site of photosynthesis in a plant cell is: 1

(a) mitochondria **(b)** nucleus **(c)** ribosome **(d)** chloroplast

2. Bile juice is secreted by: 1

(a) pancreas **(b)** liver **(c)** stomach **(d)** gall bladder

3. The breakdown of pyruvate into lactic acid occurs in: 1

(a) yeast **(b)** mitochondria with oxygen
(c) chloroplasts **(d)** muscle cells during lack of oxygen

4. A student states: “Arteries always carry oxygenated blood.” The correct response is: 1

(a) Correct
(b) Incorrect — the pulmonary artery carries deoxygenated blood
(c) Correct except the aorta
(d) Cannot be determined

5. For the opening and closing of stomata, Student A says guard cells control it; Student B says xylem controls it. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The functional unit of the kidney is: 1

(a) villus **(b)** nephron **(c)** alveolus **(d)** neuron

7. Which blood vessel carries blood from the lungs to the heart? 1

(a) pulmonary vein **(b)** aorta **(c)** pulmonary artery **(d)** vena cava

8. The enzyme present in saliva is: 1

(a) salivary amylase **(b)** lipase **(c)** pepsin **(d)** trypsin

9. Transport of food in plants takes place through: 1

(a) root hair **(b)** stomata **(c)** phloem **(d)** xylem

10. In humans, the number of chambers in the heart is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why are the walls of ventricles thicker than those of atria? 2

12. Write the balanced equation for photosynthesis. 2

13. What are the products of anaerobic respiration in yeast? 2

14. A student says that the small intestine is long in herbivores. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe the role of 3
(i) hydrochloric acid
(ii) pepsin
(iii) mucus in the stomach.

16. Draw a labelled diagram of a nephron (describe its parts) and state how urine is formed. 3

17. Explain double circulation in humans. Why is it necessary? 3

OR

Differentiate between aerobic and anaerobic respiration (three points). 3

18. How is water transported in plants? Explain the role of transpiration. 3

OR

What are the raw materials of photosynthesis and where do plants get them from? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe the process of digestion in the small intestine. 5
(ii) What are villi? State their function.
(iii) Name the enzymes of pancreatic juice.

OR

(i) Describe the human respiratory system with a labelled diagram (main parts). 5
(ii) How are the lungs designed to maximise gas exchange?

20. **Case study:** Destarched potted plants were used. One leaf was covered with black paper strips, and the plant was kept in sunlight for 6 h. The leaf was then tested with iodine. 5
(i) Why was the plant destarched?
(ii) What colour do covered and uncovered parts give?
(iii) What conclusion is drawn?
(iv) Why is the leaf boiled in alcohol?

OR

(i) Describe the structure and working of the human heart. 5
(ii) Why do fishes have a two-chambered heart?

END OF QUESTION PAPER — Total: 40 Marks

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